

Multiplying & Dividing Fractions — Error Detective Answer Key

Grade 5

Quick Answers

- | | |
|----------------------------|------------------------------------|
| 1. $\frac{8}{15}$ | 5. 8 |
| 2. $1 \cdot \frac{1}{4}$ | 6. (b) = $1 \cdot \frac{1}{6}$, — |
| 3. $\frac{15}{8}$ | 7. 10 |
| 4. (b) = $\frac{5}{4}$, — | 8. 4 |
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What is verified

6 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 6. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 5

5.NF.B – problems 1, 2, 5, 6

6.NS.A.1 – problems 3, 4, 7, 8

Difficulty 1–4 of 5 across 10 tagged checks.

Common wrong answers

4. *If they got 0.8: flipped the first fraction instead of the second*

6. *If they got 0.296: cross-multiplied: numerator times the other fraction's denominator*

1. Multiply: $\frac{2}{3} \times \frac{4}{5}$

Solution: Multiply straight across — numerator times numerator, denominator times denominator:

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$

Nothing cancels, so the product is already in simplest form.

$\frac{8}{15}$

2. Multiply, simplest form: $\frac{5}{6} \times \frac{3}{10}$

Solution: Multiply straight across, then simplify (or cancel a top number with a bottom number first — 5 with 10, 3 with 6):

$$\frac{5}{6} \times \frac{3}{10} = \frac{5 \times 3}{6 \times 10} = \frac{15}{60}$$

Dividing top and bottom by 15 gives the answer.

$\frac{1}{4}$

3. Divide: $\frac{3}{4} \div \frac{2}{5}$

Solution: Keep the first fraction, change $\div$ to $\times$, flip the *second* fraction (the divisor):

$$\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$$

As a mixed number that is $1\frac{7}{8}$.

$\frac{15}{8}$

4. Jordan's division $\frac{5}{8} \div \frac{1}{2}$ (find and fix)

Solution: (a) Jordan flipped the *first* fraction: $\frac{5}{8}$ became $\frac{8}{5}$. The keep-change-flip rule flips only the *divisor*, the fraction you are dividing by ($\frac{1}{2}$). A correct response must say that the wrong fraction was inverted.

(b) Done correctly:

$$\frac{5}{8} \div \frac{1}{2} = \frac{5}{8} \times \frac{2}{1} = \frac{10}{8}$$

which reduces to the answer.

$\frac{5}{4}$

5. Flour: 6 cups, each loaf uses $\frac{3}{4}$ cup

Solution: “How many $\frac{3}{4}$ s fit in 6?” is a division:

$$6 \div \frac{3}{4} = 6 \times \frac{4}{3} = \frac{24}{3}$$

So Mr. Diaz can make
loaves.

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6. Ava's product $\frac{2}{9} \times \frac{3}{4}$ (find and fix)

Solution: (a) Ava *cross*-multiplied (2×4 over 9×3) as if she were solving a proportion. Multiplying fractions goes straight across: top times top over bottom times bottom. A correct response must name the cross-multiply (or “multiplied diagonally”) mistake.

(b) Done correctly:

$$\frac{2}{9} \times \frac{3}{4} = \frac{2 \times 3}{9 \times 4} = \frac{6}{36}$$

which simplifies to the answer.

$\frac{1}{6}$

7. Ribbon: $\frac{5}{6}$ m cut into $\frac{1}{12}$ -m pieces

Solution: “How many $\frac{1}{12}$ s fit in $\frac{5}{6}$?” is a division:

$$\frac{5}{6} \div \frac{1}{12} = \frac{5}{6} \times \frac{12}{1} = \frac{60}{6}$$

So there are
pieces.

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8. Challenge: $2\frac{1}{2} \div \frac{5}{8}$

Solution: First rewrite the mixed number as an improper fraction: $2\frac{1}{2} = \frac{5}{2}$. Then keep-change-flip:

$$\frac{5}{2} \div \frac{5}{8} = \frac{5}{2} \times \frac{8}{5} = \frac{40}{10}$$

which is the whole number

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